

Section 4.6 Exercises

Directions: For exercises 1 - 7, solve each equation for the variable.

1. $4^{-3v-2} = 4^{-v}$

2. $64 \cdot 4^{3x} = 16$

3. $3^{2x+1} \cdot 3^x = 243$

4. $2^{-3n} \cdot \frac{1}{4} = 2^{n+2}$

5. $625 \cdot 5^{3x+3} = 125$

6. $\frac{36^{3b}}{36^{2b}} = 216^{2-b}$

7. $\left(\frac{1}{64}\right)^{3n} \cdot 8 = 2^6$

Directions: For exercises 8 - 25, solve each equation for the variable.

8. $9^{x-10} = 1$

9. $2e^{6x} = 13$

10. $e^{r+10} - 10 = -42$

11. $2 \cdot 10^{9a} = 29$

12. $-8 \cdot 10^{p+7} - 7 = -24$

13. $7e^{3n-5} + 5 = -89$

14. $e^{-3k} + 6 = 44$

15. $-5e^{9x-8} - 8 = -62$

16. $-6e^{9x+8} + 2 = -74$

17. $2^{x+1} = 5^{2x-1}$

18. $e^{2x} - e^x - 132 = 0$

19. $7e^{8x+8} - 5 = -95$

20. $10e^{8x+3} + 2 = 8$

21. $4e^{3x+3} - 7 = 53$

22. $8e^{-5x-2} - 4 = -90$

23. $3^{2x+1} = 7^{x-2}$

24. $e^{2x} - e^x - 6 = 0$

25. $3e^{3-3x} + 6 = -31$

Directions: For exercises 26 - 30, solve each equation for the variable.

26. $5\log_7 n = 10$

27. $-8\log_9 x = 16$

28. $4 + \log_2(9k) = 2$

29. $2\log(8n + 4) + 6 = 10$

30. $10 - 4\ln(9 - 8x) = 6$

Directions: For exercises 31 - 38, solve each equation for the variable.

31. $\ln(10 - 3x) = \ln(-4x)$

32. $\log_{13}(5n - 2) = \log_{13}(8 - 5n)$

33. $\log(x + 3) - \log(x) = \log(74)$

34. $\ln(-3x) = \ln(x^2 - 6x)$

35. $\log_4(6 - m) = \log_4 3m$

36. $\ln(x - 2) - \ln(x) = \ln(54)$

37. $\log_9(2n^2 - 14n) = \log_9(-45 + n^2)$

38. $\ln(x^2 - 10) + \ln(9) = \ln(10)$

Directions: For exercises 39 - 45, solve each equation for the variable.

39. $\log(x + 12) = \log(x) + \log(12)$

40. $\ln(x) + \ln(x - 3) = \ln(7x)$

41. $\log_2(7x + 6) = 3$

42. $\ln(7) + \ln(2 - 4x^2) = \ln(14)$

43. $\log_8(x + 6) - \log_8(x) = \log_8(58)$

44. $\ln(3) - \ln(3 - 3x) = \ln(4)$

45. $\log_3(3x) - \log_3(6) = \log_3(77)$

46. The population of a small town is modeled by the equation $P = 1650e^{0.5t}$ where t is measured in years. In approximately how many years will the town's population reach 20,000?

47. Atmospheric pressure P in pounds per square inch is represented by the formula $P = 14.7e^{-0.21x}$, where x is the number of miles above sea level. To the nearest foot, how high is the peak of a mountain with an atmospheric pressure of 8.369 pounds per square inch? (Hint: there are 5280 feet in a mile)